

Searching for an Exoplanet Using Pulsar Timing

Sahil Shah

Abstract: Pulsar timing is a radio method that is incredibly precise for discovering exoplanets; this is done by measuring periodic oscillations in the time of arrival (TOA) of the pulsar's pulses. In this project, our aim is to rediscover planet PSR J1719-1438 b using pulsar data. We plan to apply timing techniques with libstempo to model and analyse the periodic patterns from data collected by the Parkes radio telescope. Our preliminary results still show that we need to refine our data and model to find periodic patterns. After that we will continue to investigate properties about the binary system.

1 Scientific Background of Pulsars

1.1 Pulsars

Pulsars are highly magnetised and rapidly rotating neutron stars[1] that emit electromagnetic waves from the magnetic poles of the star. Pulsars vary in size but are typically of the magnitude of a few kilometres in diameter, similar to the size of a small city, yet they have the mass of a normal star, making them extremely dense stellar objects. Pulsar research is incredibly important to astrophysics research, as they can be considered atomic clocks due to how stable the periods of the pulses are[2].

Pulsars can be modelled as the "lighthouse" model as shown in figure 1, which depicts how radiation is expelled in the form of a rapidly rotating beacon [3] from both magnetic poles. Due to its incredibly strong magnetic field, at the magnetic poles where the field is strongest, with its high-frequency rotation speed, matter is accelerated along the field and released in the form of radio waves.

1.2 Pulsar Timing

We observe pulses when they cross our line of sight using radio telescopes. From the data received, we can measure a number of quantities, such as the system to which the pulsar belongs and the basic astrophysical characteristics (i.e. mass, luminosity, period). Since pulsar pulses are extremely regular, we can measure the small variations in time of arrival (TOA). These variations are caused by a number of factors, such as interstellar medium, solar wind, gravitational waves, or, most notably, exoplanets. The first set of exoplanets orbiting PSR B1257 + 12[5] was found this way, proving that finding exoplanets using pulsar timing is a sensitive method.

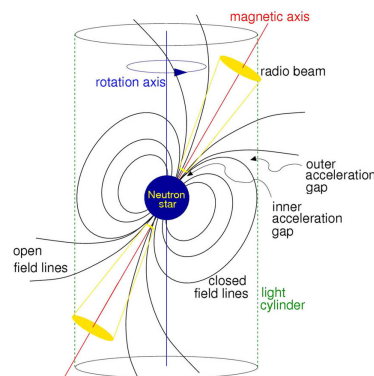


Figure 1: Model depicting the 'lighthouse' effect by B. Ya. Losovsky [4] showing the conic sections released along the magnetic axis, which is off-centre the rotation axis

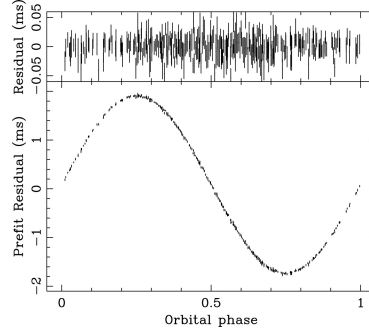


Figure 2: Pre-fit residuals for PSR 1719-1438[8]

The general idea is that the exoplanet ”tugs” the pulsar as they orbit around a common centre known as the barycentre; this causes the distance between Earth and the pulsar to vary ever so slightly. We can measure these small variations in the TOA and compare them to a model of when the pulses should arrive if there was no planet. This periodic variation is given in the form of a sinusoidal wave[6], showing the period of the exoplanet.

2 Project Aims

During this project, our aim is to reproduce the method to find exoplanets around pulsars. There are only eight confirmed planets, using pulsar timing[7]. Our aim for this project is to use timing data collected about a pulsar that has a confirmed planet and attempt to obtain characteristics about the system such as the period and mass of the planet and compare them to the true values. Looking through the exoplanet archive, we wanted to narrow our searches to a binary system that consisted only of a singular planet and a singular pulsar. From this we chose millisecond pulsar PSR J1719-1438, which has only one confirmed exoplanet[8].

PSR J1719-1438, as mentioned above, is a binary system consisting of only a millisecond pulsar and the planet. From Bailes’ paper, the system shows that the pulsar has a period of 5.7 milliseconds and the planet, PSR J1719-1438b, is considered to be an exotic planet with a $P = 2.2hrs$. This ”exotic” planet is theorised to be the remains of a carbon white dwarf that is roughly the size of Jupiter[8]. This system was observed using the Parkes 64-meter radio telescope located in Australia.

Our main objective for the coming weeks is, ”Can we, using pulsar timing, rediscover the exotic exoplanet orbiting PSR J1719-1438?”. This objective is a relatively open one; to prove the planet exists would mean that we found the period of the planet through plotting the periodic oscillations of the residuals. If we were to have more time, we could also include calculations to find the mass of the planet and other characteristics of the pulsar binary system.

3 Data and Method

To achieve our main objective of finding the period, we are using data provided by Bailes in his paper, from the Parkes telescope, as supporting evidence[8]. He provided two files in the form of ”.par” and ”.tim” files. These files are very common when observing pulsars and using timing models. ”.par” files provide the parameters about the pulsar, such as the period, ascension, declination, and proper motion. The ”.tim” files provide the TOA of the pulses. Using this, we can plot the residuals of each pulse detected against time as shown in figure 3. A residual is the small variation in time from when the pulse is supposed to arrive.

To plot our data and manipulate it, we plan to use the advised software known as Tempo2[9]. However, tempo2 is written in FORTRAN and C, which we do not have any experience with, so we will use a Python wrapper known as libstempo. Using the wrapper, we plan to refine the

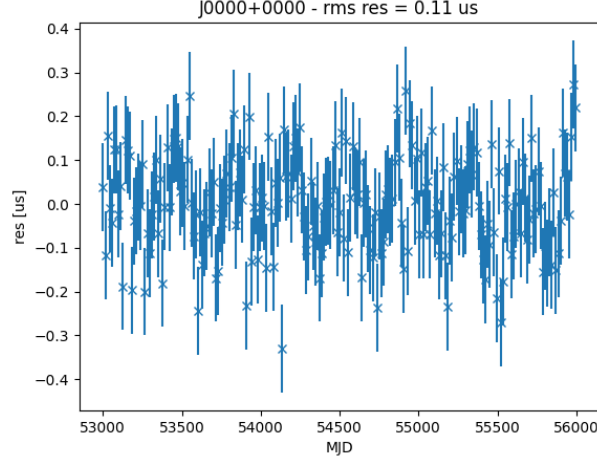


Figure 3: Test plot of a simulated pulsar of $P = 15\text{ms}$, showing residuals against time

data and return plots that will eventually show us a plot of residuals against orbital phase, similar to that shown in figure 2. Using the parameters about PSR J1719-1438 we can also figure out characteristics about the pulsar, such as its surface magnetic field and energy loss of the system, as well as find out characteristics about the companion planet, such as the mass.

To work out the mass of the pulsar, m_p , and companion, m_c , we can put constraints on them, and obtain a mass function[1]:

$$f_{mass} = \frac{4\pi^2}{G} \frac{x^3}{P_b^2} = \frac{(m_c \sin i)^3}{(m_p + m_c)^2} \quad (1)$$

where G is the gravitational constant, i is the angle between the plane of orbit and the plane of the sky, x is the projected semi-major orbital axis, and P_b is the orbital period of the binary system. Given these parameters, we can calculate the maximum mass of the companion by substituting $i = 90^\circ$, assuming the mass of the pulsar is known.

Furthermore, we can work out more parameters about the pulsar just using the period, such as its surface magnetic field (equation 3), and energy loss (equation 2)[10].

$$\dot{E} = 4\pi^2 I \dot{P} P^{-3} \approx 3.95 \times 10^{31} \text{ergs}^{-1} \left(\frac{\dot{P}}{10^{-15}} \right) \left(\frac{P}{s} \right)^3 \quad (2)$$

$$B_s = 3.2 \times 10^{19} G \sqrt{P \dot{P}} \approx 10^{12} G \left(\frac{\dot{P}}{10^{-15}} \right)^{0.5} \left(\frac{P}{s} \right)^{0.5} \quad (3)$$

where P and $\dot{P}$ are the period and period derivative, respectively. These parameters could easily be derived from the data given by the '.par' file. The file contains parameters $f0$ and $f1$, which can be converted to the period using the relationship $f = \frac{1}{P}$.

4 Preliminary Results

Over the last few weeks, we have started to conduct our research and calculations. To start, we wanted to learn how to use the libstempo wrapper for Tempo2, so we followed a tutorial[11] that introduced the package and basic commands. This tutorial made us simulate a pulsar as shown in figure 3, with a period of 15ms . We worked on making our own pulsar by making our own ".par" file and using the time file function to create a TOA file. The tutorial also used equations 2 and 3 to calculate $\dot{E}$ and B_s for this simulated pulsar.

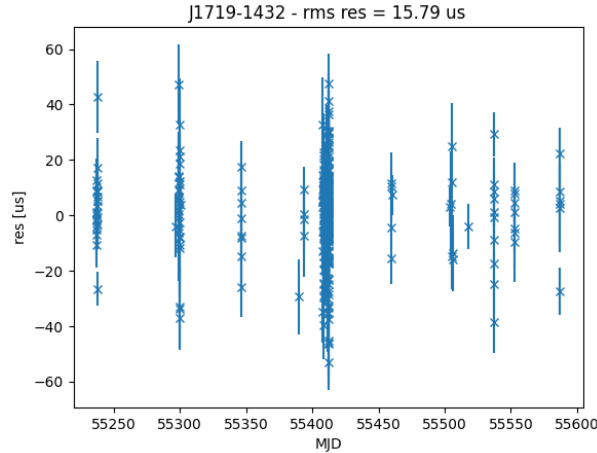


Figure 4: Residual Plot of actual data of PSR 1719 - 1438

The latest progress we have made is producing a similar residual graph to the simulated pulsar, as shown by figure 4. This plot differs a lot compared to the simulated; this is most likely due to the fact that the radio telescope was not recording for the full length of 400 MJD (Modified Julian Date, a convention used by astrophysicists, which refers to day 0 being on the midnight of November 17th, 1858 [12]). Since we know the period of the planet we are working towards is around two hours, we know that we must set our x-axis resolution to be smaller, specifically around the dense, thick area in the middle of the graph, which seems to have a lot more data points, and narrow our search for periodic patterns.

5 Conclusion

Using pulsar timing is a scarce method for finding exoplanets, due to the extreme conditions left behind by the formation of neutron stars. In our project, we hope to be able to replicate the pulsar timing techniques for PSR 1719-1438 to obtain numerous characteristics about not just the planet but the pulsar as well. We hope to at least find the period of the exotic exoplanet using data from the Parkes telescope and compare it to true values published by Bailes. At the end, we want to leave extra time to find some derived characteristics, such as the surface magnetic field of the pulsar.

6 AI declaration

I have not used AI to help me write this report, nor to aid with the project in any way (e.g. research, coding and help).

References

- [1] Duncan R Lorimer. Binary and millisecond pulsars. *Living reviews in relativity*, 11:1–90, 2008.
- [2] JM Cordes, M Kramer, TJW Lazio, BW Stappers, DC Backer, and S Johnston. Pulsars as tools for fundamental physics & astrophysics. *New Astronomy Reviews*, 48(11-12):1413–1438, 2004.
- [3] HO De Oliveira, RM Marinho, and NS Magalhaes. Stellar structures and the enigma of pulsars rotation frequency decay. In *Journal of Physics: Conference Series*, volume 630, page 012059. IOP Publishing, 2015.
- [4] B. Losovsky. The specifics of pulsar radio emission. *Open Astronomy*, 31:189–197, 05 2022.
- [5] Aleksander Wolszczan and Dail A Frail. A planetary system around the millisecond pulsar psr1257+ 12. *Nature*, 355(6356):145–147, 1992.
- [6] EA Behrens, SM Ransom, DR Madison, Z Arzoumanian, K Crowter, ME DeCesar, PB Demorest, T Dolch, JA Ellis, RD Ferdman, et al. The nanograv 11 yr data set: Constraints on planetary masses around 45 millisecond pulsars. *The Astrophysical Journal Letters*, 893(1):L8, 2020.

- [7] Nasa Exoplanet Archive for Pulsar Timing - https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=PS&constraint=default_flag>0&constraint=disc_method%20like%20%27%25pul%25%27.
- [8] Matthew Bailes, Stuart D Bates, Varun Bhalerao, ND Ramesh Bhat, Marta Burgay, Sarah Burke-Spolaor, Nichi D’Amico, Simon Johnston, Michael J Keith, Michael Kramer, et al. Transformation of a star into a planet in a millisecond pulsar binary. *Science*, 333(6050):1717–1720, 2011.
- [9] GB Hobbs, RT Edwards, and RN Manchester. Tempo2, a new pulsar-timing package–i. an overview. *Monthly Notices of the Royal Astronomical Society*, 369(2):655–672, 2006.
- [10] D.R. Lorimer and M. Kramer. *Handbook of pulsar astronomy*, volume 4. Cambridge university press, 2005.
- [11] Libstempo Tutorial of a Simulated Pulsar https://github.com/VIPER-Research-Group/gw-school-2022-materials/blob/master/Week_1/Day_2/timing_tutorial/timing_libstempo_intro.ipynb.
- [12] Peter Meyer. Julian day numbers. *Hermetic Systems*. http://www.hermetic.ch/cal_stud/jdn.htm, pages 491–499, 2002.